

# Machine Vision ( Computer Vision )

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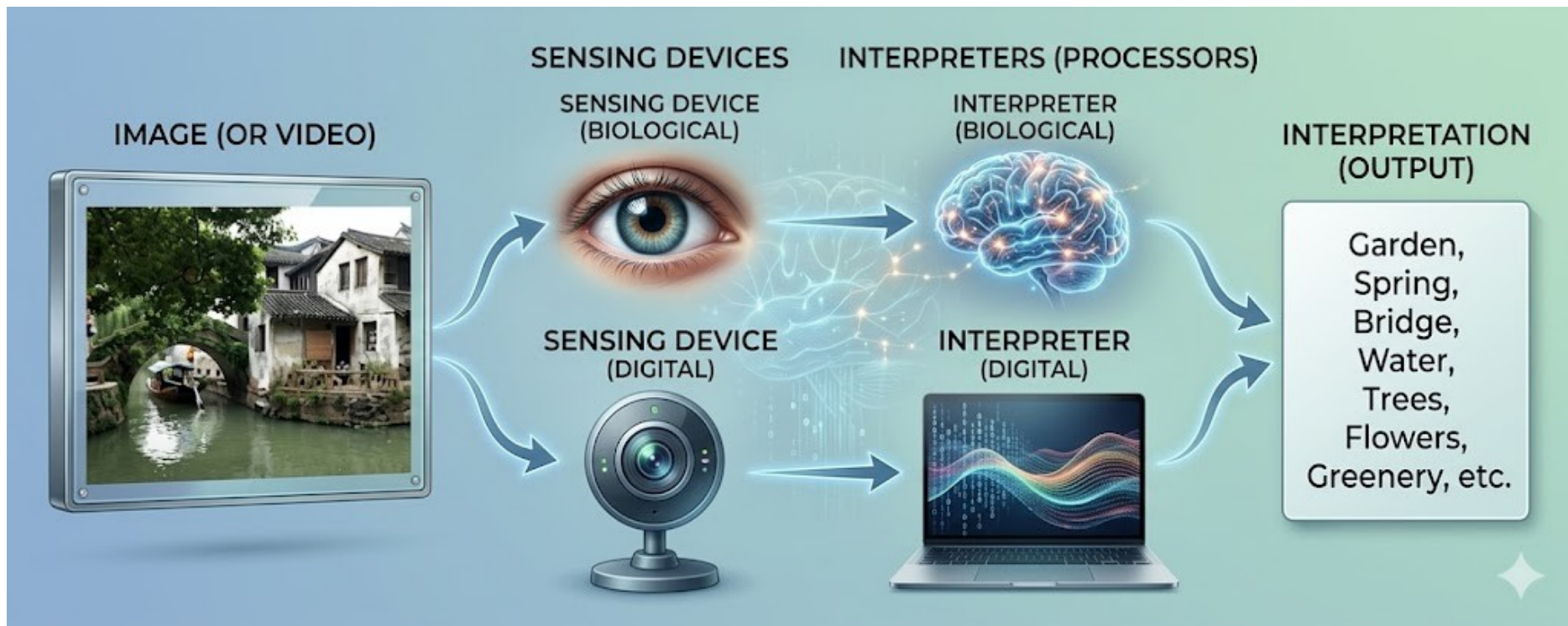
Chen Peng

School of Artificial Intelligence

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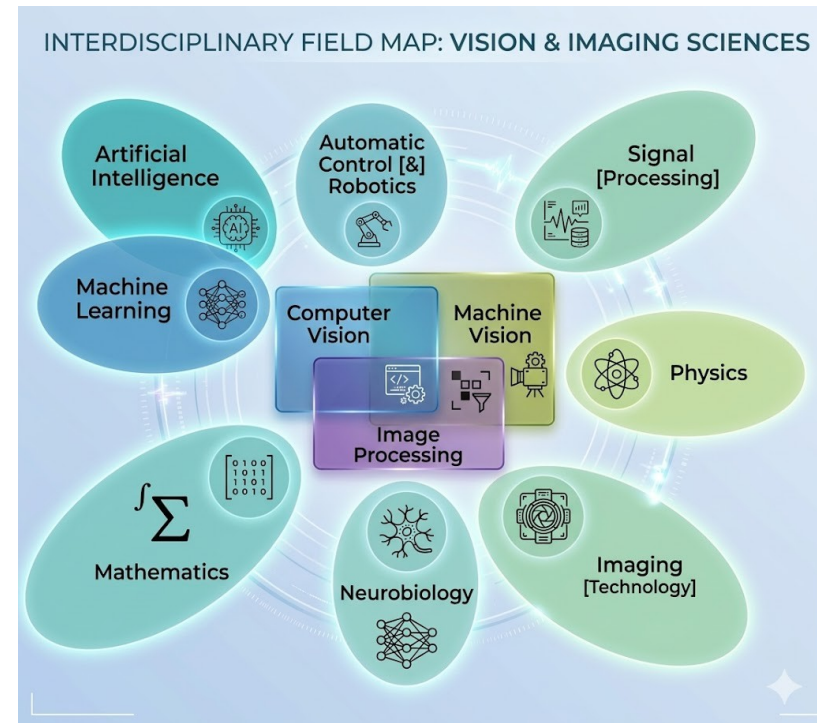


# What is Computer Vision



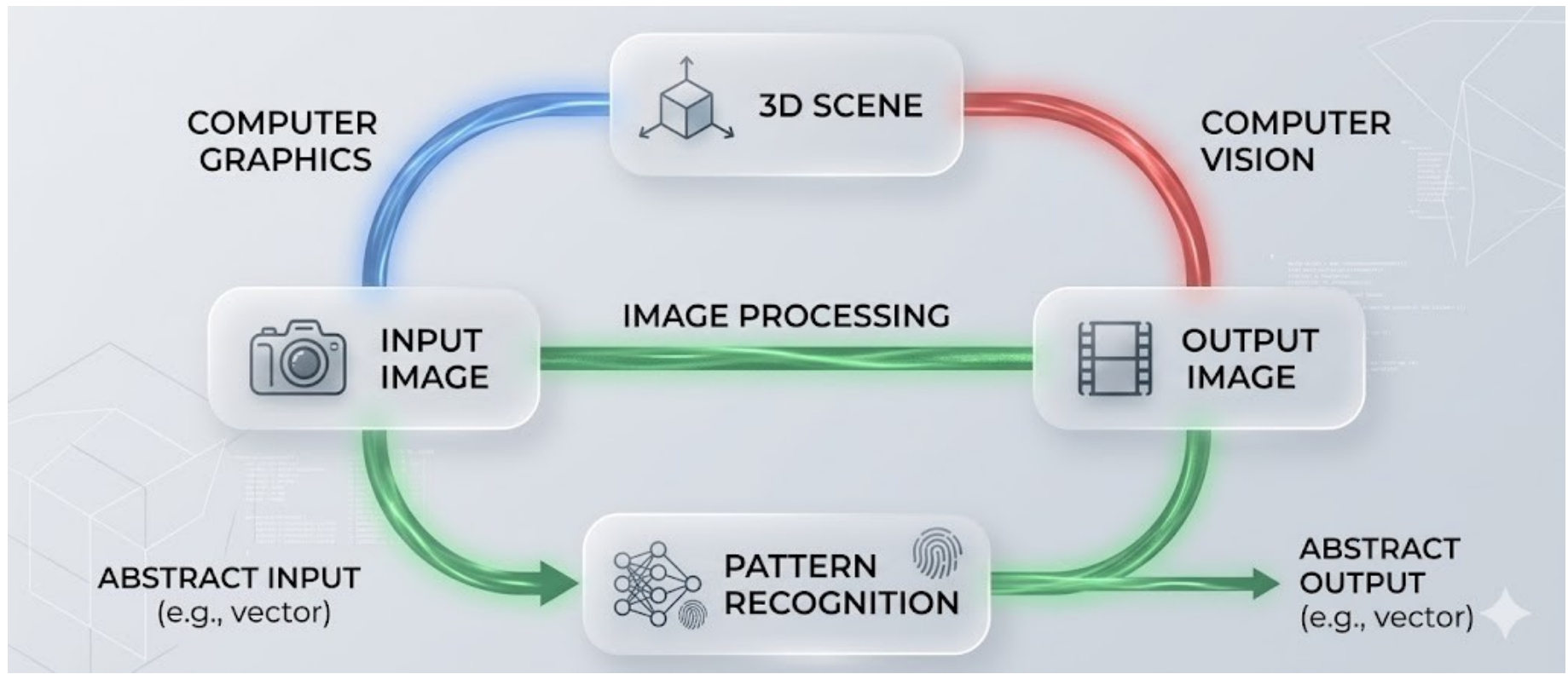
# What is Computer Vision

- **Computer Vision (CV)**, as the name suggests, is a discipline that "teaches" computers how to "see" the world.
- Against the backdrop of the surging popularity of machine learning, computer vision stands alongside **Natural Language Processing (NLP)** and **Speech Recognition** as one of the three major hotspots in the field of machine learning.
- The methods and technologies of computer vision have **partial overlaps with many disciplines**, including: artificial intelligence, digital image processing, machine learning, deep learning, pattern recognition, probabilistic graphical models, scientific computing, and a series of mathematical computations.



# Machine Vision vs Computer Vision

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# Machine Vision vs Computer Vision

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- Machine Vision (MV) leans more towards the engineering application of computer vision technology, enabling the automatic acquisition and analysis of specific images to control corresponding behaviors.
- A machine vision system is essentially a system capable of automatically **acquiring** one or more images of a target object, **processing, analyzing,** and **measuring** various feature quantities of the acquired images, and **providing qualitative analysis and quantitative interpretation of the measurements**, thereby gaining certain insights about the target object and **making corresponding decisions**.

# Machine Vision vs Computer Vision

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- The purpose of Machine Vision is to better provide technical support for the **manufacturing industry to improve product quality and production efficiency.**
- The functions of Machine Vision include: **object localization, feature detection, defect judgment, target recognition, counting, and motion tracking.**
- Precisely because machine vision systems can rapidly acquire large amounts of information, are easy to process automatically, and are readily integrated with design information and processing control information, **they are widely used in fields such as condition monitoring, finished product inspection, and quality control in modern automated production processes.**

# Machine Vision vs Computer Vision

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- Computer Vision (CV) employs a combination of image processing, pattern recognition, and artificial intelligence technologies, focusing on the computer analysis of one or multiple images. Specifically, Computer Vision provides the theoretical and algorithmic foundation for image and scene analysis for Machine Vision, while Machine Vision provides the sensor models, system construction, and implementation means for the realization of Computer Vision.
- The purpose of researching Computer Vision is to bring "light" to computers based on the characteristics of human vision, enabling them to better replace humans in work or to accomplish tasks that humans cannot, thereby greatly improving production efficiency while continuously enhancing people's quality of life.



# Machine Vision vs Computer Vision

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| MACHINE VISION         | INDUSTRIAL VISION                                                                                                                   | COMPUTER VISION                                                                                                          |
|------------------------|-------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| Application Areas      | Smart Manufacturing                                                                                                                 | Future consumption, service, and smart living areas.                                                                     |
| Functional Goals       | Primarily addresses repetitive labor tasks that require human eyes for workpiece positioning, measurement, inspection, etc.         | Endows intelligent robots with vision, enabling the recognition and judgment of external position and image information. |
| Hardware Requirements  | High requirements; necessitates filtering industrial cameras based on frame rate, resolution, and other indices according to needs. | Except for specific cases, most requirements for cameras or image sensors are low.                                       |
| Algorithm Requirements | Often focuses on improving precision.                                                                                               | More complex; focuses on using mathematical logic or deep learning for object calibration and recognition.               |
| Industry Maturity      | Relatively high; already widely applied in measurement and inspection in industries like semiconductors and packaging.              | Overall, still in the preliminary exploration stage, with startups emerging continuously. ✨                              |



# Why vision is so hard

- Bridging the "Semantic Gap" Establishing the mapping from pixels to semantics



What we saw

|   |   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|---|
| 0 | 3 | 2 | 5 | 4 | 7 | 6 | 9 | 8 |
| 3 | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 |
| 2 | 1 | 0 | 3 | 2 | 5 | 4 | 7 | 6 |
| 5 | 2 | 3 | 0 | 1 | 2 | 3 | 4 | 5 |
| 4 | 3 | 2 | 1 | 0 | 3 | 2 | 5 | 4 |
| 7 | 4 | 5 | 2 | 3 | 0 | 1 | 2 | 3 |
| 6 | 5 | 4 | 3 | 2 | 1 | 0 | 3 | 2 |
| 9 | 6 | 7 | 4 | 5 | 2 | 3 | 0 | 1 |
| 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |

What the computer saw

# Why vision is so hard

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- 1981: Nobel Prize in Physiology or Medicine



Hubel & Wiesel

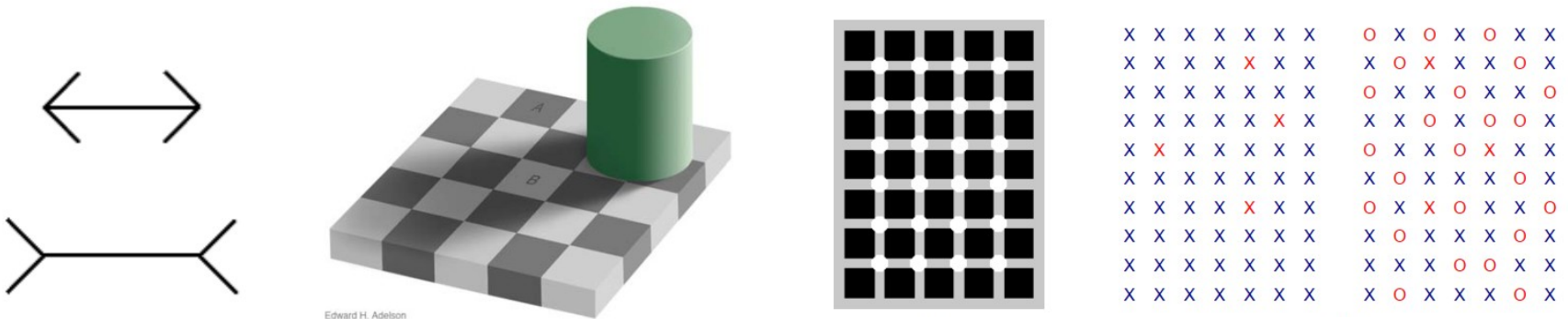
Dr. Hubel said:

There has been a myth that the brain cannot understand itself. It is compared to a man trying to lift himself by his own bootstraps.

We feel that is nonsense. The brain can be studied just as the kidney can.

# Why vision is so hard

- Part of the reason is that it is an inverse problem, where we attempt to recover some unknowns to provide a complete solution in the presence of insufficient information.
- We must resort to physics-based and probability-based models to disambiguate the potential solutions.



# Why vision is so hard

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- In the field of computer vision, we attempt to perform the reverse process, that is, to describe the world we see from one or more images, such as shape, illumination, and color distribution.
- Amazingly, humans and animals can accomplish this effortlessly, whereas computer vision algorithms are prone to errors. People who have not worked in this field often underestimate the difficulty of the problem.
- It was initially believed that the cognitive aspects of intelligence (logical reasoning and planning) were inherently more difficult than the perceptual aspects.

# The application of CV

## AI Vision

Following the breakthroughs in deep learning, AI vision has entered of scaled commercial implementation, rapidly empowering sectors like Security, Pan-Finance, Retail, Healthcare, Autonomous Driving, Pan-Industrial, and Pan-Agricultural. It replaces human eyes with machines for object and face recognition, and identity verification, significantly boosting human operational efficiency.

### Security

#### Static Face Comparison

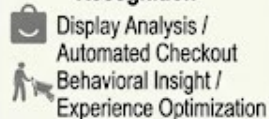


#### Dynamic Video Analysis



### Retail

#### Product/Behavior Recognition



Display Analysis /  
Automated Checkout  
Behavioral Insight /  
Experience Optimization

#### Supply Chain Digitalization



Data flow across  
people, goods, and locations,  
enabling digital supply  
chain operations

### Autonomous Driving



Delivers AI vision functions  
for demands like environment  
perception, vehicle  
localization, and mapping

### Pan-Finance



Identity  
Authentication

Account Login  
Remote Account Opening  
Risk Control  
Face-Scan Payment  
Identity Verification  
.....

### Healthcare



Image Recognition,  
Lesion Analysis



Semantic  
Segmentation of  
Lesion Areas



Target Localization,  
Detection, and  
Screening



3D Reconstruction  
& Surgical Planning

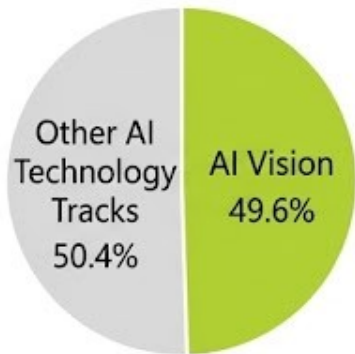
### Pan-Industrial, Pan-Agricultural

Product Quality & Patrol  
Inspection  
Safety Management  
Recognition  
Industrial Process Automation  
(Vision for robots)

Livestock/Crop Health  
Monitoring  
Product Information  
Management  
Agricultural Process Automation  
(Vision for robots)

# The application of CV

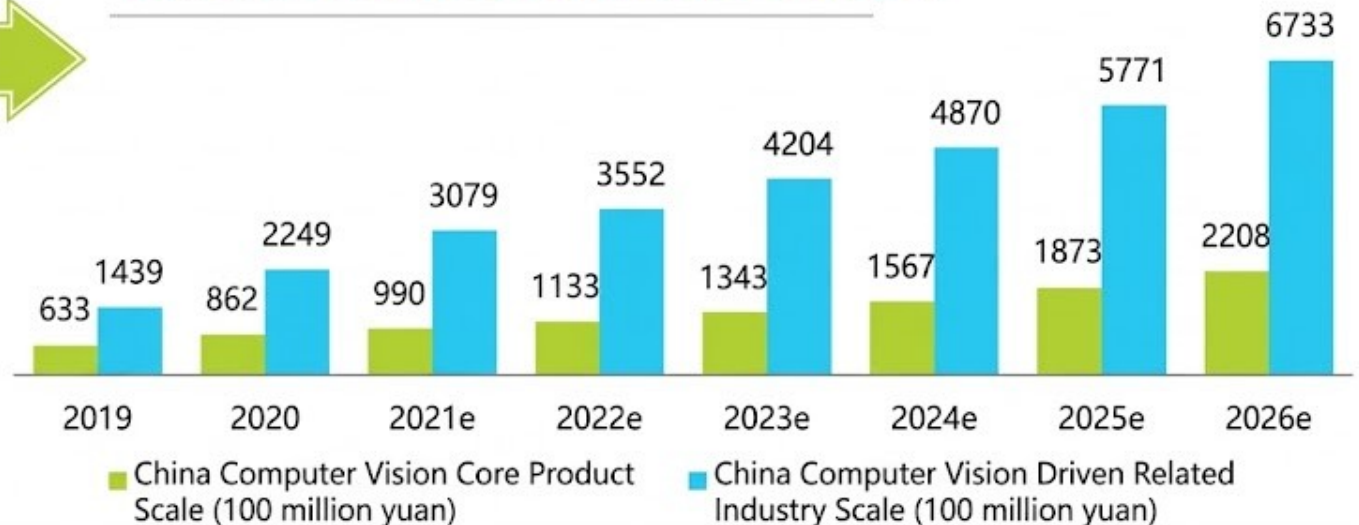
## China AI Vision Core Products and Driven Related Industry Scale, 2021-2026



China's AI vision market accounted for 49.6% of the overall AI market in 2021

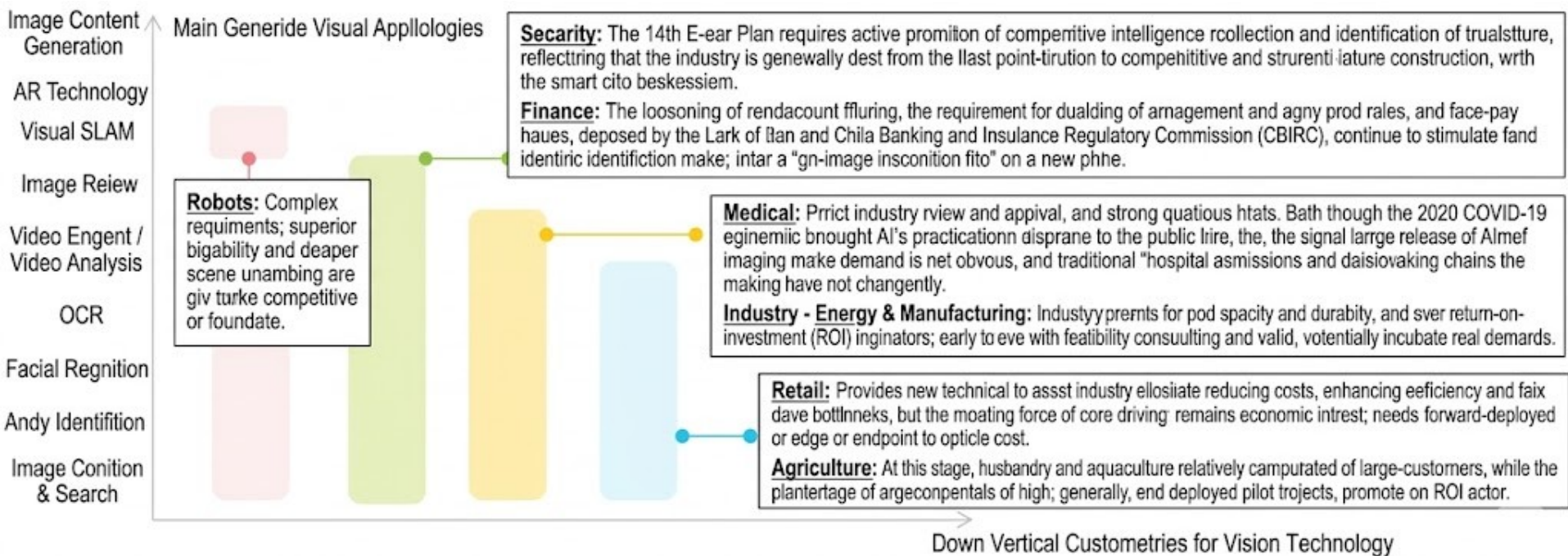
AI vision core product CAGR, 2021-2026: 17.4%

AI vision driven related industry CAGR, 2021-2026: 16.9%





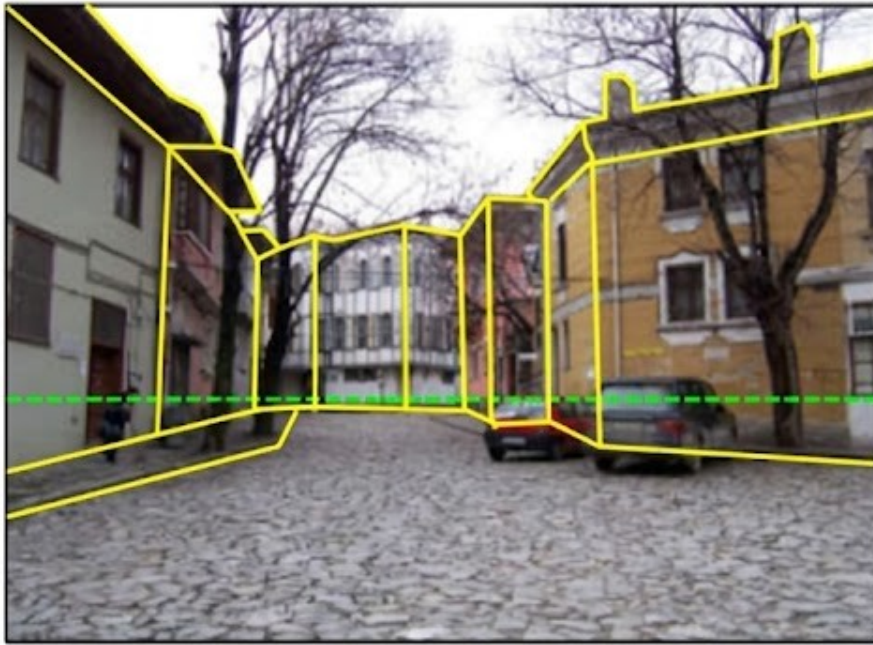
# The application of CV





# The topics of CV

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**Structural Information of 3D Scene**



**Semantic Information**

# The topics of CV

## AI Vision

### Semantic Perception

#### Classification

Image classification is the foundation for object detection, image segmentation, object tracking, behavior analysis, face recognition, and other high-level vision tasks. It differentiates image categories based on the image's semantic information.



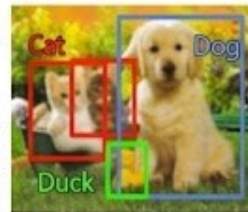
#### Detection

Object detection outputs a Bounding Box and label for each individual target object in an image, using the box to denote the object's location and providing the object's category for classification.



#### Segmentation

Image segmentation addresses the question of 'which object or scene each pixel belongs to' simplify or alter the image representation, making it easier to understand and analyze. It is divided into semantic segmentation and instance segmentation.



#### Recognition

- Object Recognition: Performing object recognition on content like license plates and text.
- Recognition related to the human body, such as face recognition, fingerprint recognition, action recognition, iris recognition, etc.



Taking action recognition as an example

### Location & Tracking

#### Localization

Using computer vision technology to find the position of a specific target object within an image, i.e., localization.

#### Tracking

Target tracking aims to track objects that are continuously moving over time, using consecutive video frames as input.



### Geometric Attributes

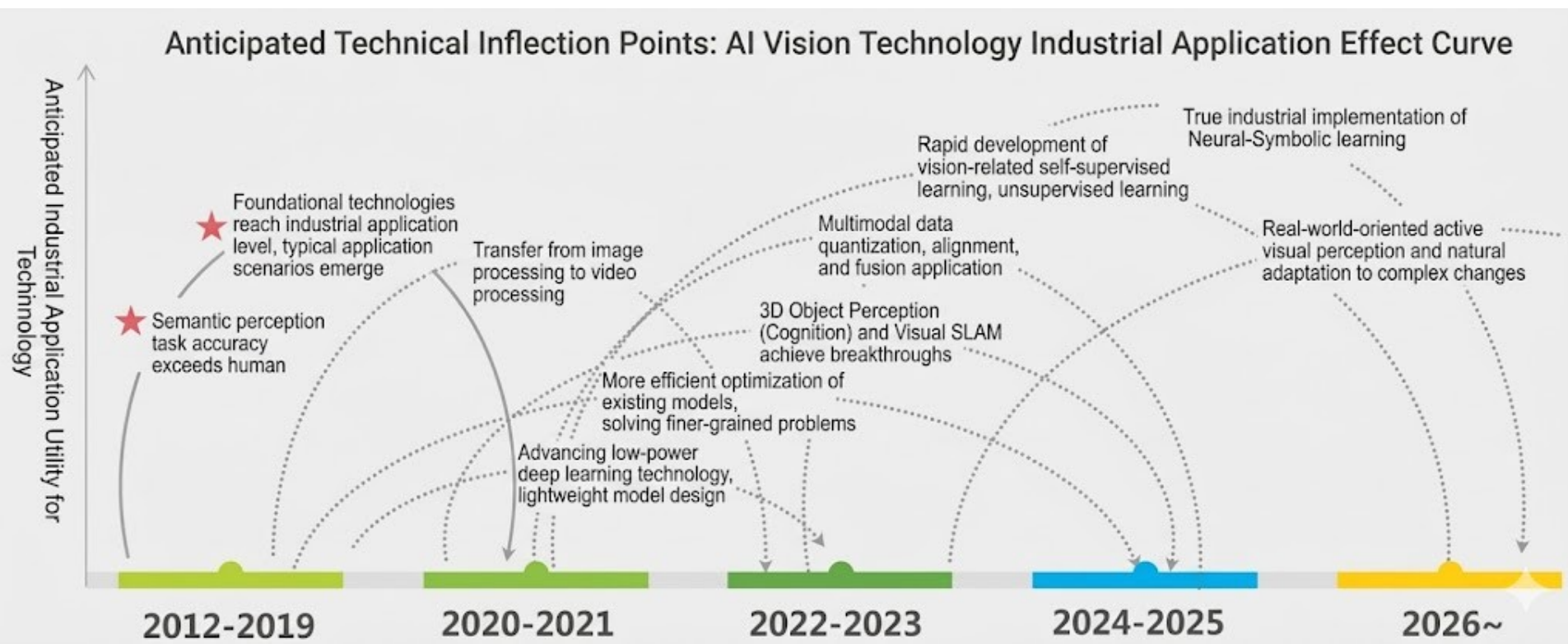
#### 3D Modeling

#### Visual SLAM (Simultaneous Localization and Mapping)

#### Augmented Reality

Involves 3D target perception and reconstruction, which are closely related to applications in robotics, AR, etc., focusing on the geometric attributes of AI vision.

# The topics of CV



# The topics of CV

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- With the Transformer model, which has shone brightly in the field of Natural Language Processing (NLP), being applied to the field of Computer Vision (CV), hybrid model architectures combining it with CNNs are gradually becoming a key research direction for vision tasks, aiming to reduce the complexity of model structures and improve scalability and training efficiency.
- Vision technology still awaits breakthroughs in adapting to the **3D world**, breaking through the limitations of **relying on labeled data input, reducing computational energy consumption, multimodal information fusion and analysis, combining with knowledge and common sense to solve high-level problems**, and **active perception and adaptation to complex changes**.
- However, "technical homogenization" does not imply "algorithmic homogenization"; the engineering capability of AI vision algorithms remains the touchstone for the industrial implementation of technology.



# The history of CV

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## Embryonic Stage: Two-Dimensional to Three-Dimensional 1950s-1960s

- AI vision began in the 1950s with statistical pattern recognition, when work was mainly focused on two-dimensional image analysis and recognition; in the 1960s, research aimed at three-dimensional vision understanding was pioneered.

## Enlightenment Stage: Becoming an Independent Discipline 1970s-1980s

- The publication of David Marr's book "Vision" in 1982 marked the establishment of AI vision as an independent discipline. Marr proposed the "Marr's Computational Vision" which consists of three levels: computational theory, representation and algorithm, and implementation, providing an important theoretical framework for vision research.

## Exploration Stage: Era of Feature Engineering 1990s – Early 21st Century

- In the 1990s, vision theories based on multi-view geometry developed rapidly. Statistical learning methods triggered a significant shift, with statistical learning methods like support vector machines applied in AI vision, and feature object recognition began to take focus. Early in the 21st century, image feature engineering began, with the appearance of high-quality datasets that were truly annotated.

## Boom Stage: Era of Neural Networks After 2012

- In 2012, "Deep Convolutional Neural Networks" (CNN) appeared, performing ImageNet dataset. Their feature is to integrate to combine traditional feature engineering and model learning into one, meaning that feature design can be conducted during the learning process, which dramatically reduced identification error rates. From then on, deep learning popular in the vision and AI fields, related applications blossomed, and AI vision entered a period of robust development.

# The history of CV

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- The Origin of Computer Vision: The "Summer Vision Project" by an MIT Undergraduate

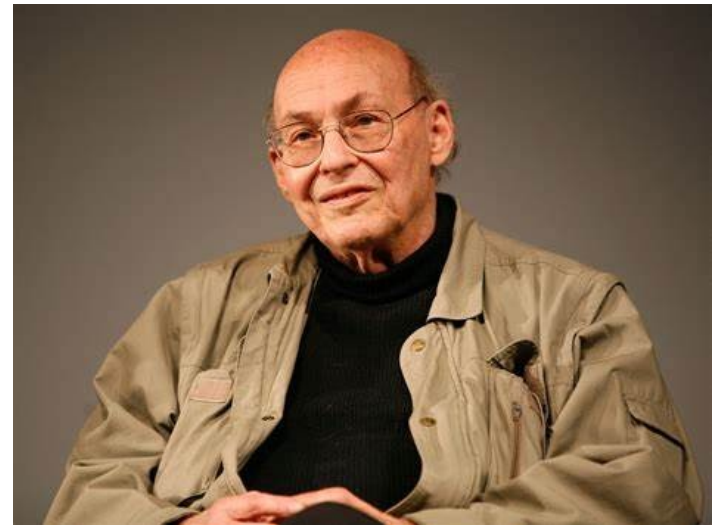
MASSACHUSETTS INSTITUTE OF TECHNOLOGY  
PROJECT MAC

Artificial Intelligence Group  
Vision Memo. No. 100.

July 7, 1966

THE SUMMER VISION PROJECT  
Seymour Papert

The summer vision project is an attempt to use our summer workers effectively in the construction of a significant part of a visual system. The particular task was chosen partly because it can be segmented into sub-problems which will allow individuals to work independently and yet participate in the construction of a system complex enough to be a real landmark in the development of "pattern recognition".



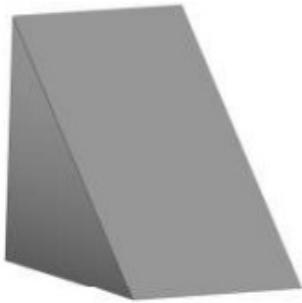


# The history of CV

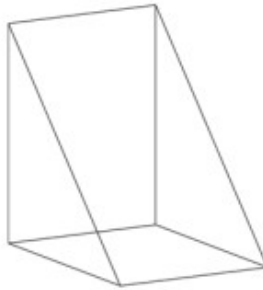
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- The Block World

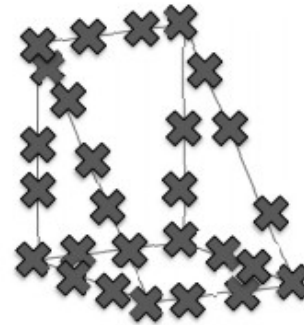
拉力.罗伯茨,1963



(a) Original picture



(b) Differentiated picture



(c) Feature points selected

# The history of CV

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- Pioneer of the Field: David Marr (1945-1980), known as the Father of Computer Vision.
- The Marr Prize is a prestigious award in the field of computer vision, presented by the committee of the International Conference on Computer Vision (ICCV). It is recognized as one of the highest honors in computer vision research.



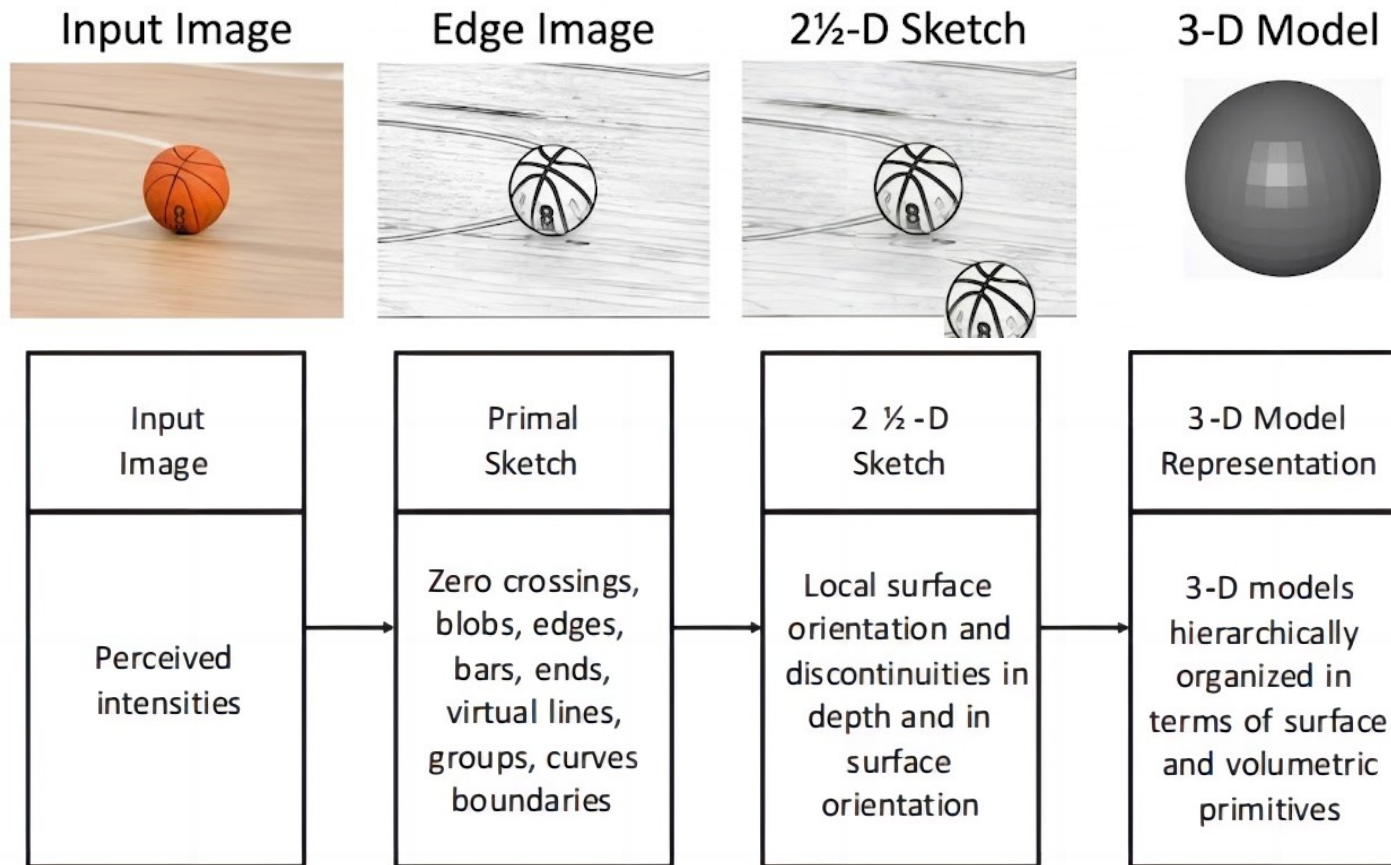
# The history of CV

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- **Computational Theory:** What is the goal of the computation? What are the known constraints or constraints that can be imposed on the problem?
- **Representation and Algorithms:** How are the input, output, and intermediate information represented? What algorithms are used to compute the desired results?
- **Hardware Implementation:** How are the representation and algorithms mapped onto physical hardware, such as biological visual systems or specialized silicon chips? Conversely, how can hardware constraints be used to guide the selection of representations and algorithms? As the use of graphics chips and multi-core architectures in computer vision continues to grow, this question has once again become quite important.

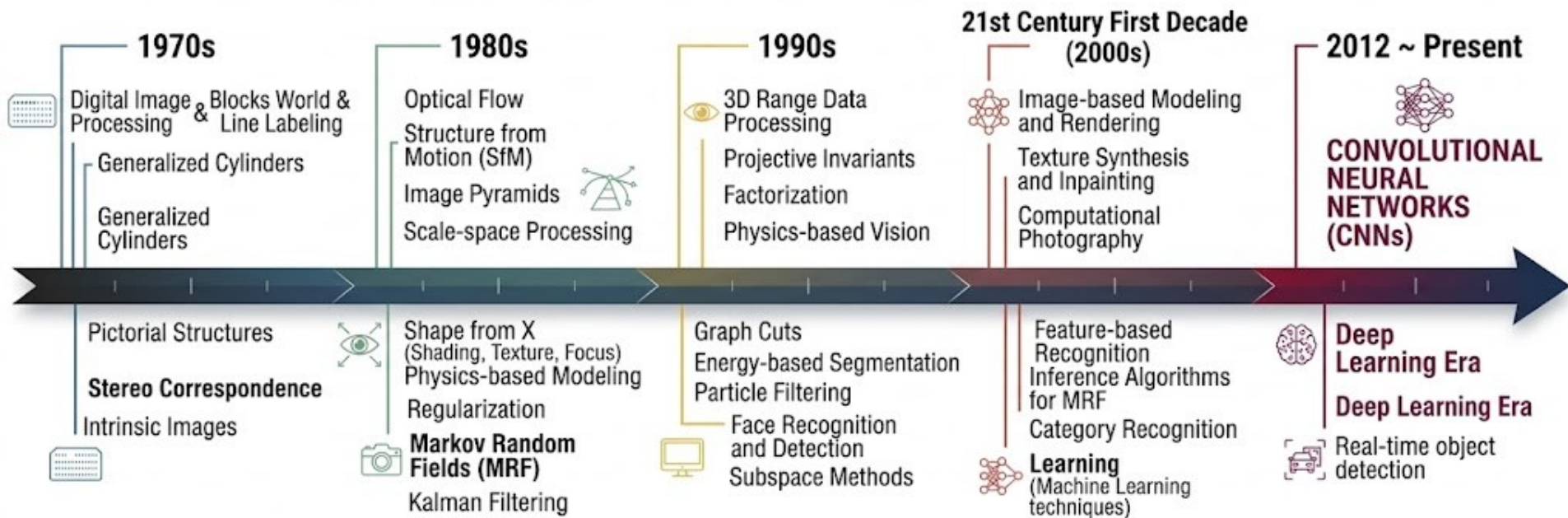
# The history of CV

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# The history of CV

## A HISTORY OF COMPUTER VISION: KEY DEVELOPMENTS





# The history of CV

## AlexNet

ImageNet 2012

| Model                 | Top-1 (val) | Top-5 (val) | Top-5 (test) |
|-----------------------|-------------|-------------|--------------|
| <i>SIFT + FVs [7]</i> | —           | —           | 26.2%        |
| 1 CNN                 | 40.7%       | 18.2%       | —            |
| 5 CNNs                | 38.1%       | 16.4%       | <b>16.4%</b> |
| 1 CNN*                | 39.0%       | 16.6%       | —            |
| 7 CNNs*               | 36.7%       | 15.4%       | <b>15.3%</b> |

error: traditional methods 26.2% → CNN 15.3%

8 Layers: 5 C + 3 FC

60 million parameters & 650,000 neurons

Deep Learning + Big Data + Big GPU

Algorithm

Big data

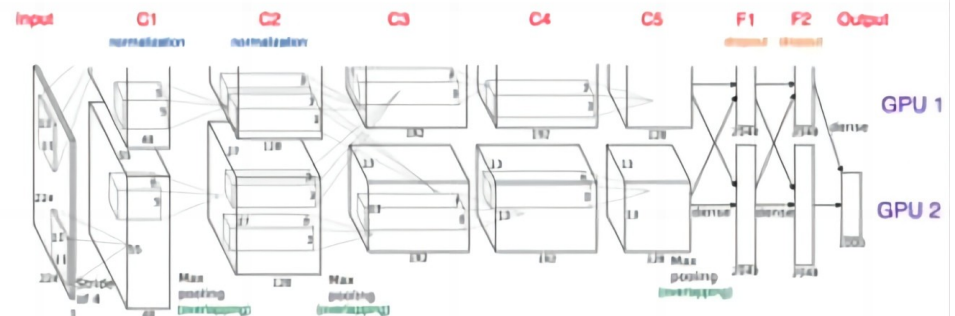
Compute

## ImageNet Classification with Deep Convolutional Neural Networks

Alex Krizhevsky  
University of Toronto  
kriz@cs.utoronto.ca

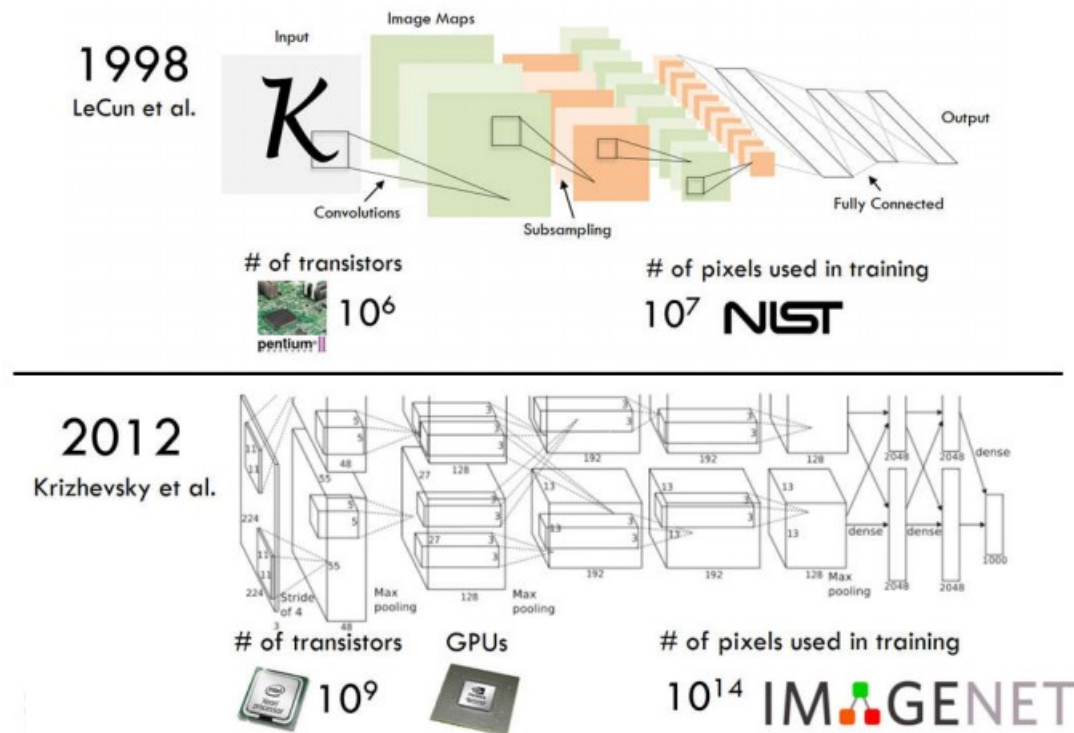
Ilya Sutskever  
University of Toronto  
ilya@cs.utoronto.ca

Geoffrey E. Hinton  
University of Toronto  
hinton@cs.utoronto.ca



# The history of CV

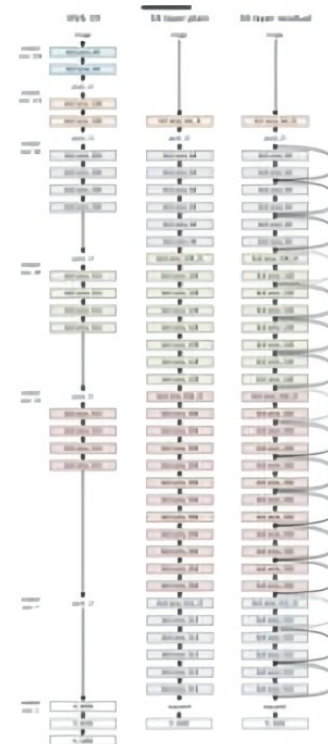
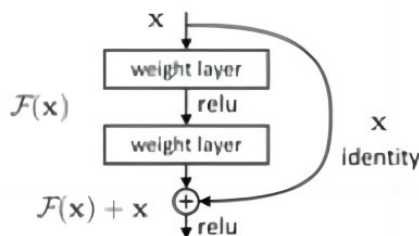
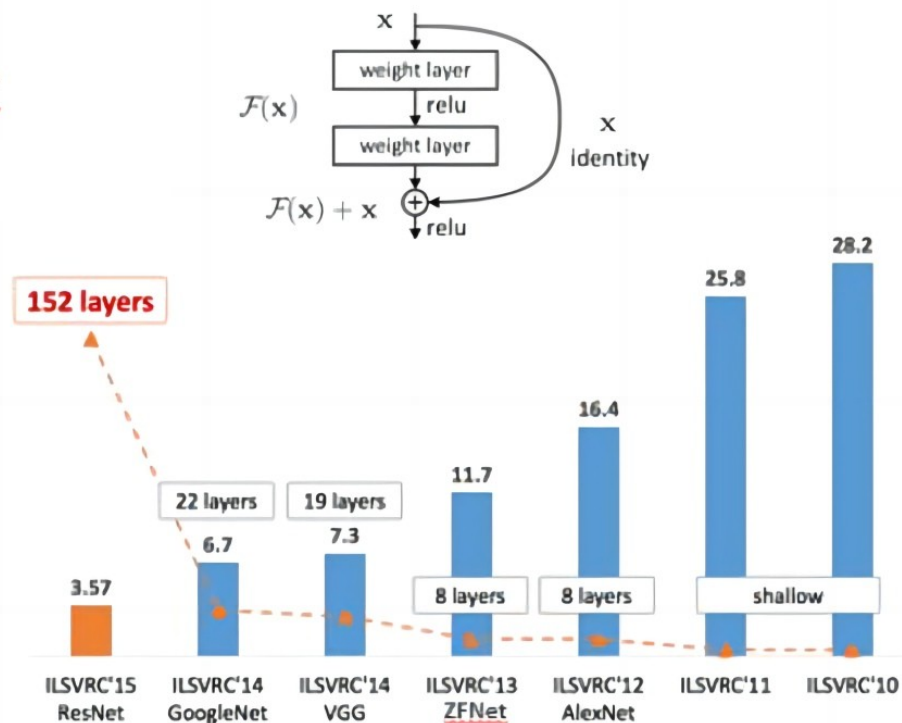
- Convolutional neural networks first appeared in 1998!



# The history of CV

2015 ResNet

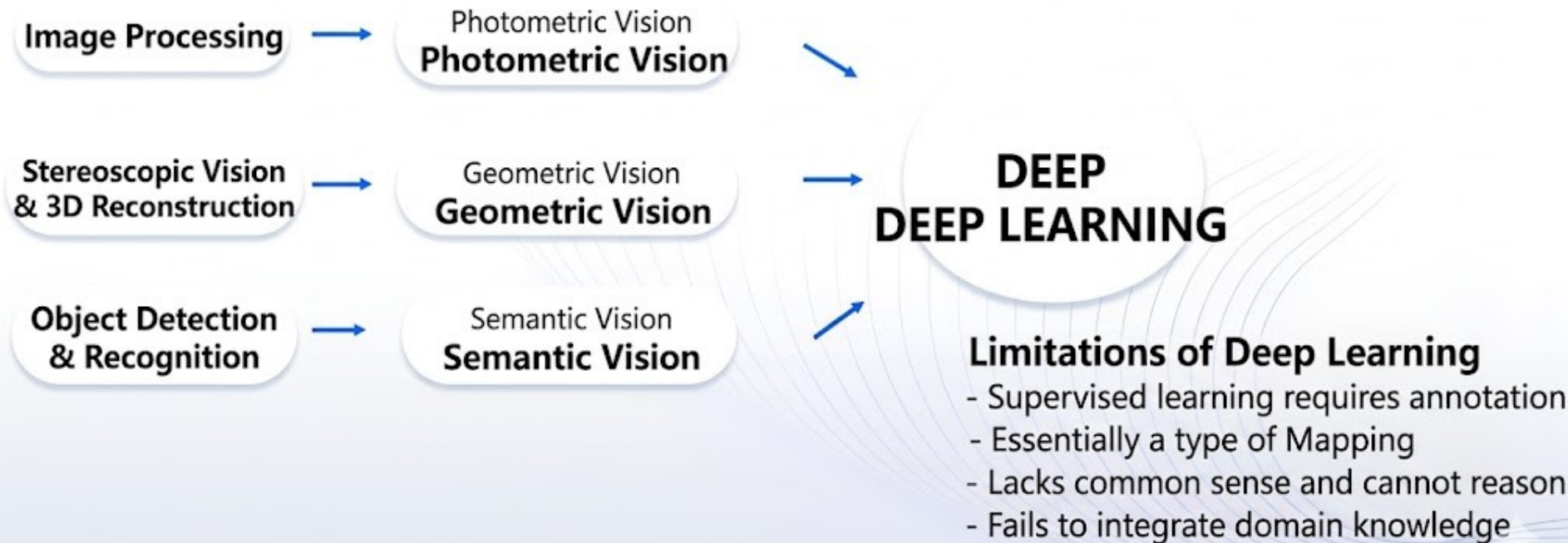
Beat human  
benchmark of  
5.1% errors



He, K; Zhang, Xu; Ren, S; Sun, J. "Deep Residual Learning for Image Recognition". arXiv:1512.03385, 2015.

# The history of CV

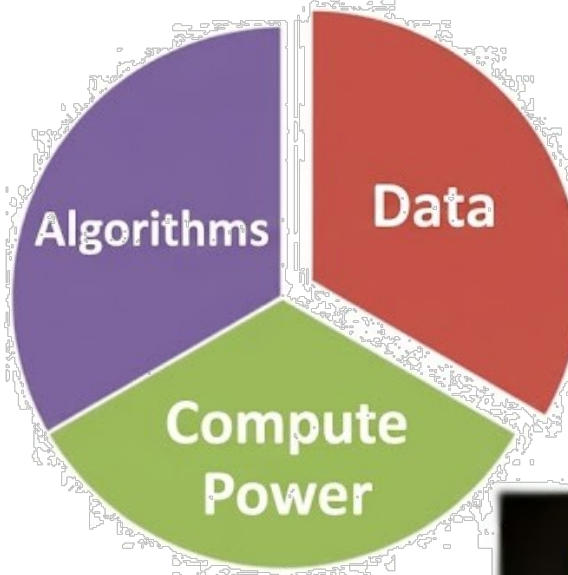
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# The history of CV

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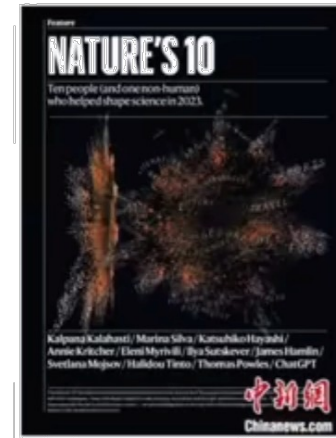
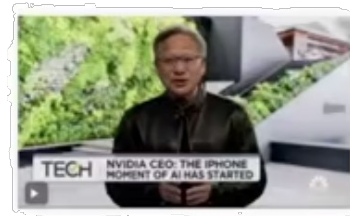
- Three Elements of Deep Learning





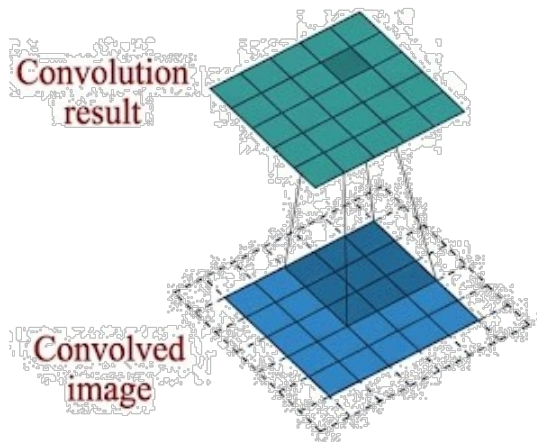
# The history of CV

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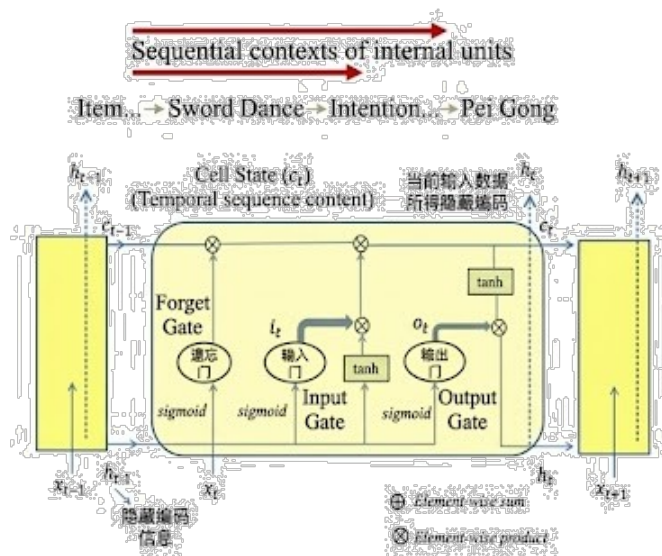


- On January 9, 2007, Steve Jobs unveiled the first-generation iPhone, seamlessly integrating an iPod, a phone, and an internet communications device, which propelled the mobile internet into its golden age of development. Today, large language models are bringing about a major transformation in numerous production and lifestyle patterns in human society.
- In February 2023, NVIDIA founder and CEO Jensen Huang proposed that with the emergence of large models represented by ChatGPT, we have entered the "iPhone moment of AI." This viewpoint has been widely recognized and republished by media outlets such as Fortune and The Wall Street Journal.
- Nature magazine's list of the Top 10 People of 2023 (Nature's 10), in addition to selecting ten individuals from major global scientific events as per convention, also featured a non-human "scene-stealer" on the list—the artificial intelligence (AI) tool ChatGPT.

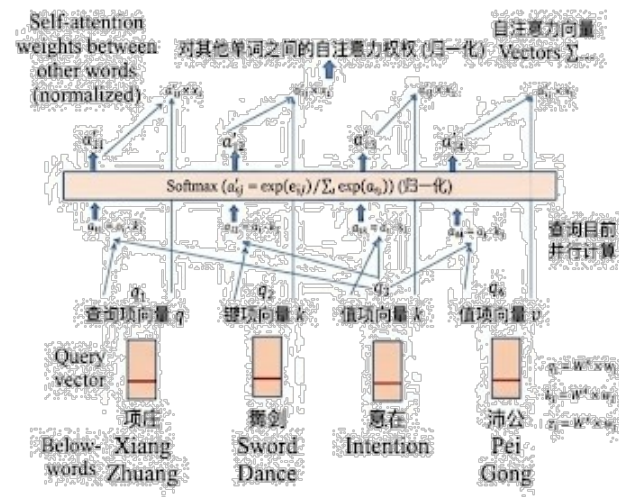
# The history of CV



**Convolutional Neural Network**  
Local spatial dependencies between pixel points

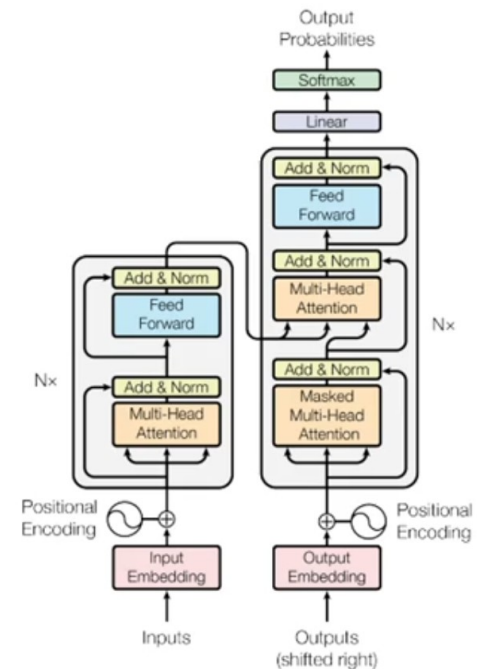
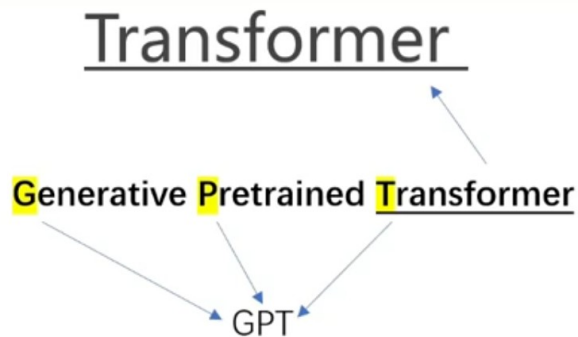


**Recurrent Neural Network (e.g., LSTM)**  
Temporal sequence dependency of sequential data



**Self-Attention Mechanism**  
Global co-occurrence relationships

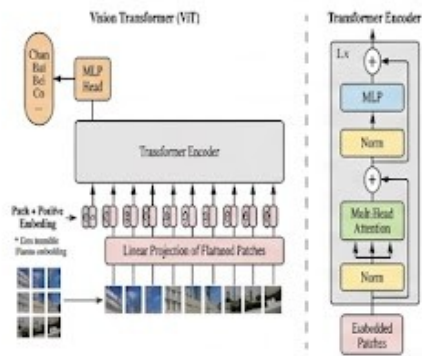
# The history of CV



2017-06

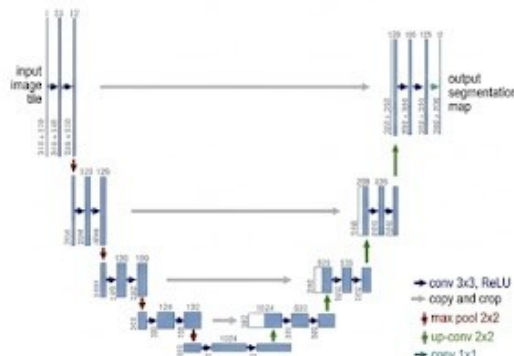
# The history of CV

## Patch-level, from language to vision



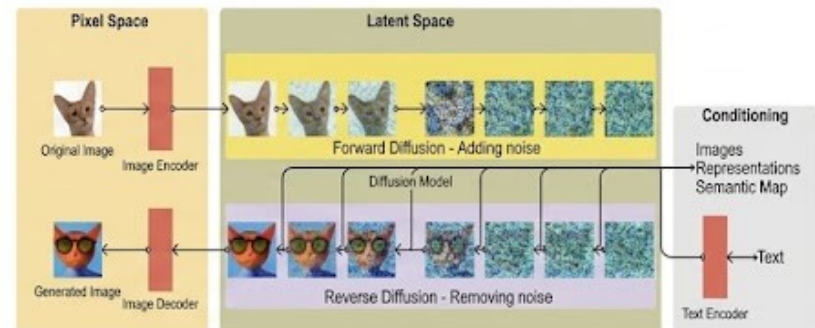
## Vision Transformer (ViT)

**Object-level,**  
from patches to objects



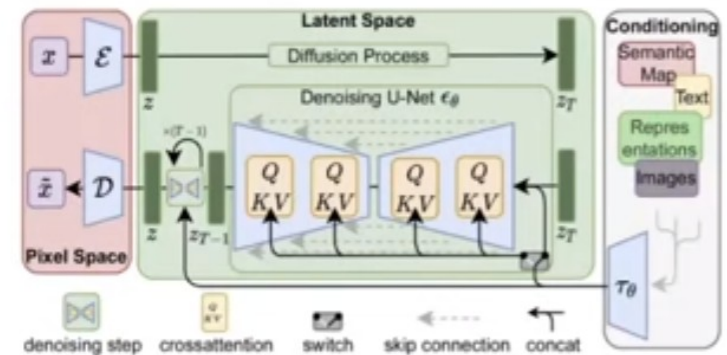
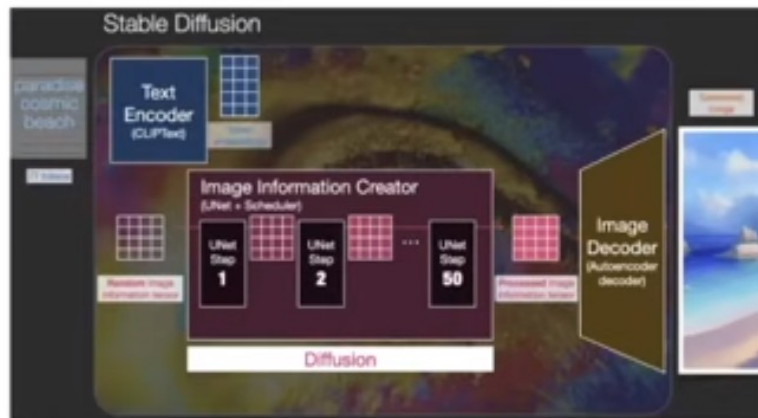
## U-Net

from pixel-space to latent space



A diffusion model in the latent space:  
knowing not just ‘that’ it works,  
but ‘how’ and ‘why’ it works

# The history of CV



- CLIP Text is used for text encoding, the U-Net combined with a Scheduler is utilized to progressively process or diffuse information transformed into the latent space, and the AutoEncoder Decoder (VAE: Variational AutoEncoder) decodes the computational results from the latent space into an image.



# The history of CV

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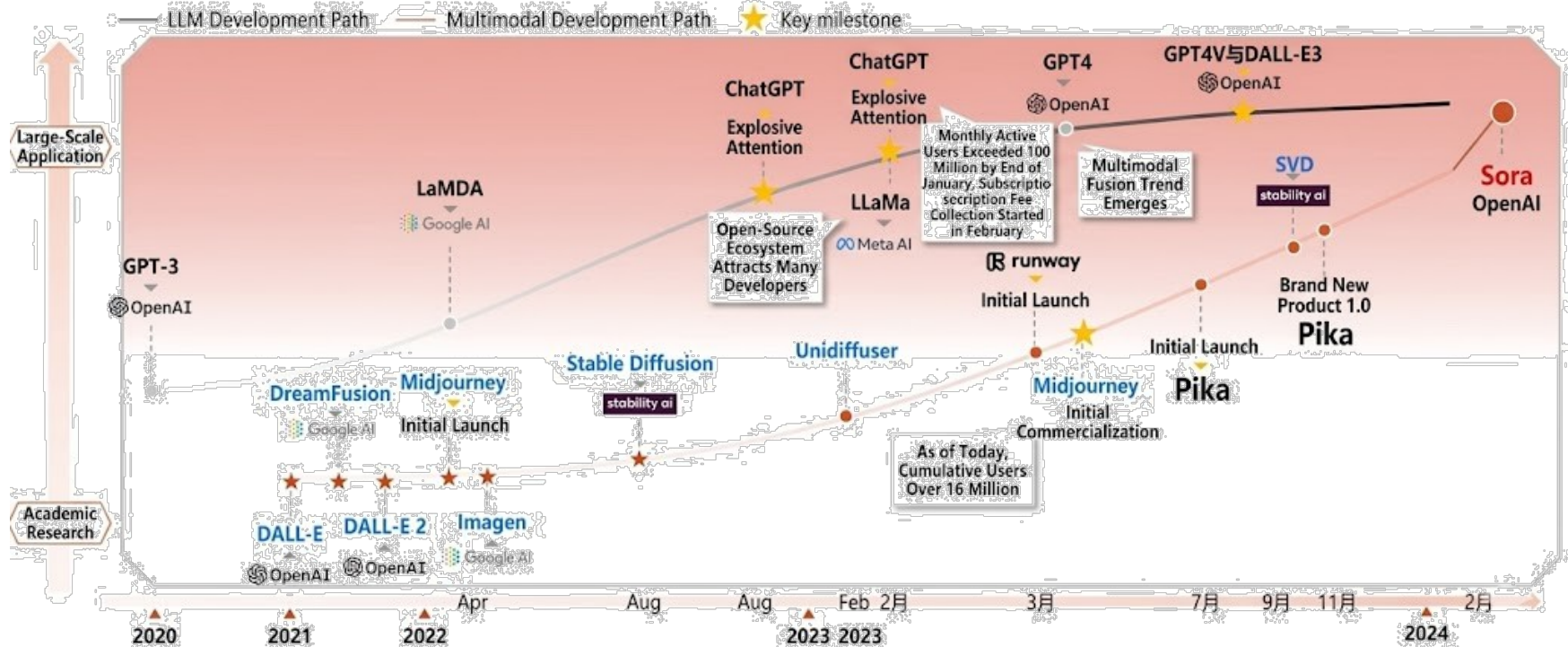
## ◆ Prompt

*[ A stylish woman walks down a Tokyo street filled with warm glowing neon and animated city signage. She wears a black leather jacket, a long red dress, and black boots, and carries a black purse. She wears sunglasses and red lipstick. She walks confidently and casually. The street is damp and reflective, creating a mirror effect of the colorful lights. Many pedestrians walk about.. ]*



# The history of CV

## Development Paths of Multimodal Large Models



# The history of CV

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**Multimodal Fusion  
(like CLIPText)**

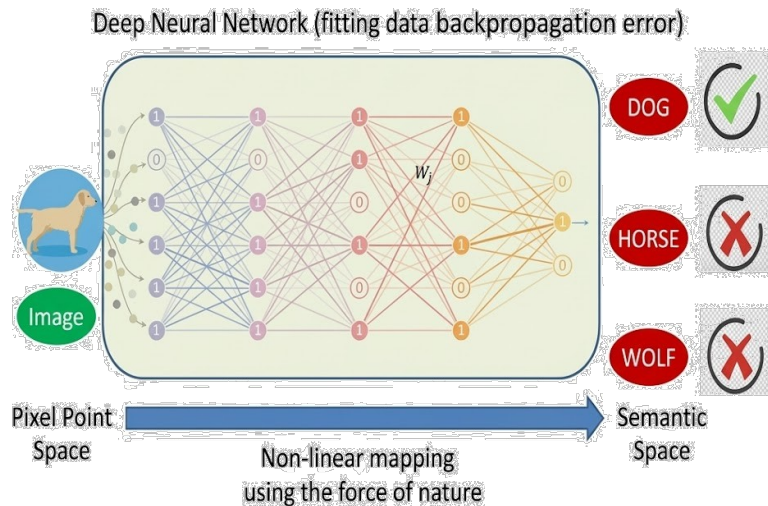
**Global Attention  
(transformer)**

Latent Space Diffusion Model  
**(latent diffusion)**

Scale up driven integration

- Rich Sutton, the pioneer of reinforcement learning in the field of artificial intelligence, published an article in 2019 titled "The Bitter Lesson." In it, he argued that "looking back on the history of AI development over the past 70 years, finding ways to leverage larger-scale computing power has always been the most efficient approach."
- It is precisely under the philosophy of continuously increasing model scale that OpenAI has placed extreme emphasis on the engineering of algorithms and the algorithmic thinking of engineering. By building a team structure and computational infrastructure where engineering and algorithms are tightly integrated, they have realized core products like ChatGPT and Sora.
- Of course, this approach of forming synthetic capabilities through constant model expansion—a feat akin to "nothing special, just practiced skill"—must have a ceiling. This is because "simplifying the complex" and achieving "great skill that appears unrefined" represent the original aspiration of advancing from "machine learning" to "learning machines."

# The history of CV



- Richard Feynman mentioned in his Lectures on Physics that in complex systems such as biology, anthropology, or economics, there is rarely a concise mathematical theory that can rival the numerical precision of mathematical physics theories. The reason lies in the fact that "they are too complex, and our minds are limited," a concept known as the Feynman Limit.
- Classical scientific theories adhere to the principle of parsimony (such as Occam's Razor), focusing on the description of principles but failing to characterize high-dimensional complex systems, thus encountering the "Feynman Limit." Data-driven machine learning, due to its function approximation capabilities, excels at discovering patterns in complex systems from a microscopic perspective and fitting high-dimensional complex systems using statistical methods. This is hailed as the power of emergence in neural network models. Emergence remains an open question.

# The history of CV

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## **The Four Inherent Defects of Large Models**

**Defect 1: Excessive Consumption of Data and Computing Power.** The parameter count of large models has reached the trillion level, with training data volume and computational consumption scaling proportionally to the number of parameters.

**Defect 2: Catastrophic Forgetting.** Training on a new task can impair performance on previously learned tasks. Furthermore, during problem-solving, these models cannot retain memory of processed data or scenarios. For instance, in autonomous driving, while the human brain retains memory of road conditions, the system must recalculate everything from scratch each time, consuming vast amounts of energy in the process.

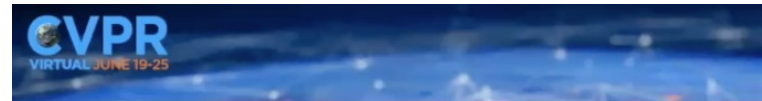
**Defect 3: Weak Logical Reasoning in Black-Box Models.** Large models lack the "divide and conquer" capability. They perform poorly when handling complex problems requiring logical or numerical reasoning, and they struggle to generalize or draw analogies (infer similar cases from one instance).

**Defect 4: Lack of Self-Awareness Regarding Errors.** Large models do not know when they are wrong, nor do they understand why they are wrong; consequently, they are incapable of correcting their mistakes. Taking the example of GPT-4 solving an arithmetic problem: it provided an incorrect answer when asked to multiply two random numbers between 1381 and 1453. Even after the error was pointed out, it could not identify whether the cause lay in the corpus, the training process, or other factors, making correction impossible.



# CV in China

- The School of Information Science and Technology at USTC has won the 2021 Marr Prize, the highest honor for computer vision papers!
- Among the total of 21,036 submitting authors, a staggering 8,203 were from China, accounting for approximately 40%. This figure far exceeds the number of authors from the United States, South Korea, and the United Kingdom.



## Marr Prize

### Swin Transformer: Hierarchical Vision Transformer using Shifted Windows

Ze Liu (USTC), Yutong Lin (Xi'an Jiaotong University),  
Yue Cao (Microsoft Research), Han Hu (Microsoft Research Asia),  
Yixuan Wei (Tsinghua University), Zheng Zhang (MSRA, Huazhong University of  
Science and Technology), Stephen Lin (Microsoft Research),  
Baining Guo (MSR Asia)

Session 8 (A/B)

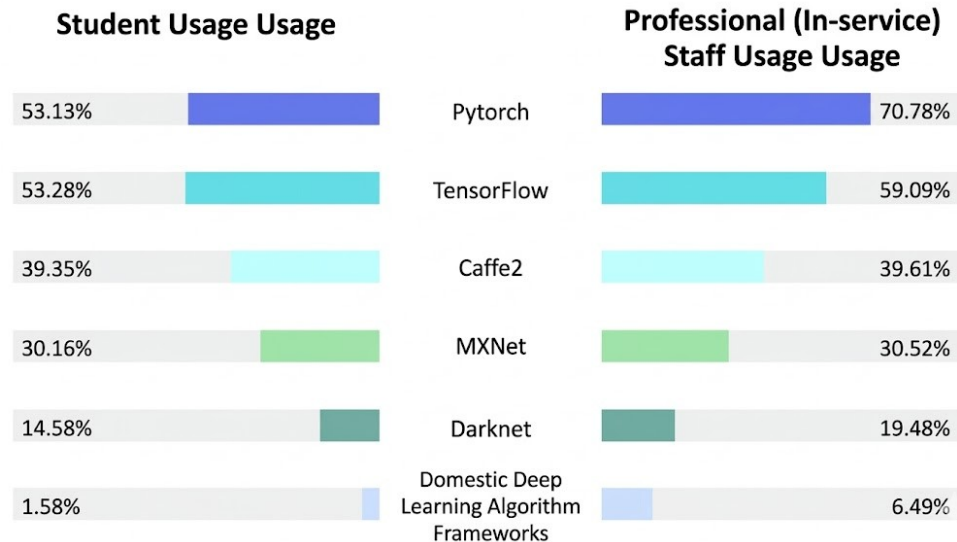
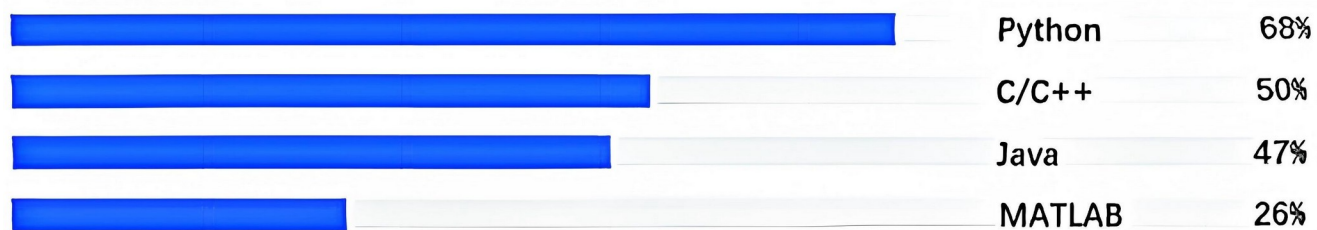
2021 ICCV OCTOBER 11-17  
VIRTUAL

## Program statistics

| Nb of authors (of submitted papers) | 21036 |
|-------------------------------------|-------|
| ... of which China                  | 8203  |
| ... of which US                     | 4628  |
| ... of which Korea                  | 1062  |
| ... of which UK                     | 655   |
| ... of which Germany                | 574   |
| ... of which Canada                 | 517   |
| ... of which Australia              | 462   |
| ... of which India                  | 429   |
| ... of which Hong Kong              | 406   |
| ... of which Singapore              | 376   |
| ... of which Japan                  | 345   |
| ... of which France                 | 325   |
| ... of which Switzerland            | 311   |
| ... of which Israel                 | 243   |

# CV in China

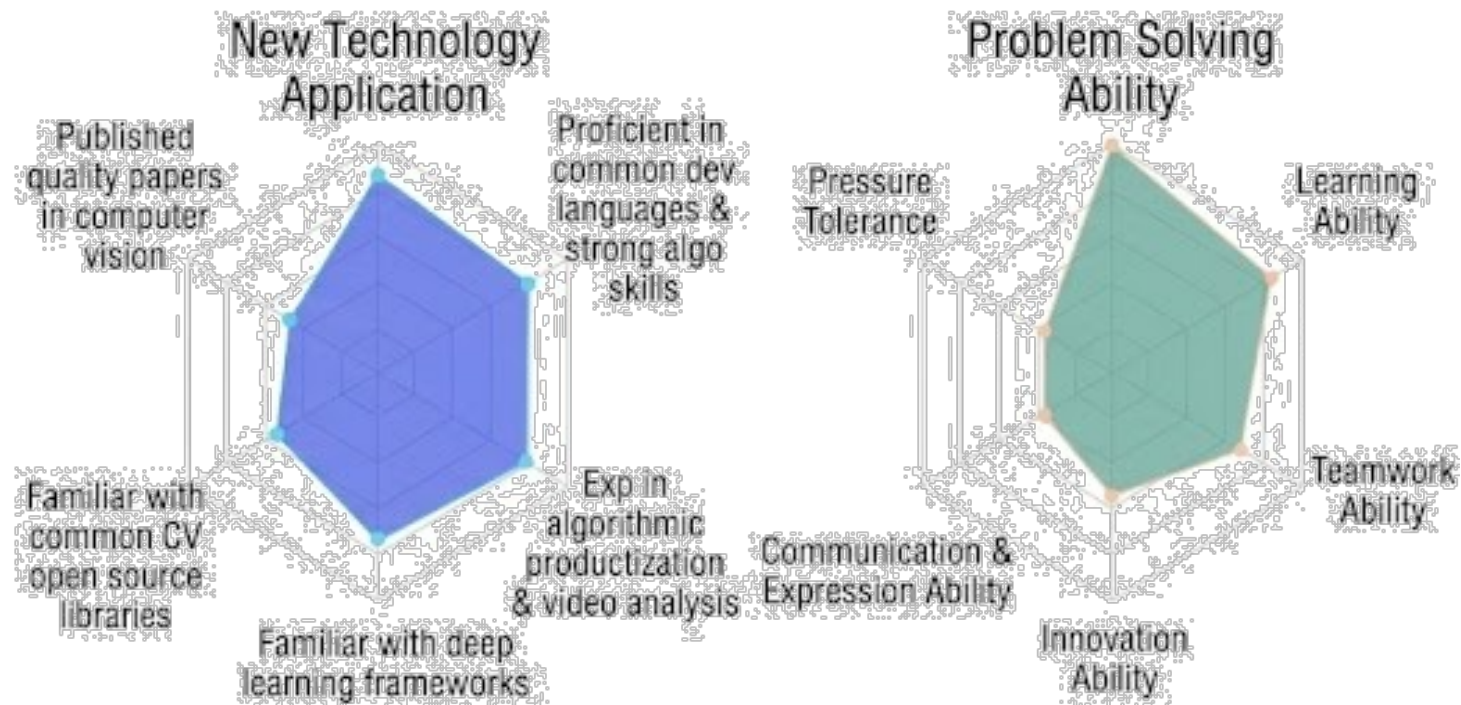
- In terms of development habits, Chinese computer vision talents most commonly use Pytorch and TensorFlow.



# CV in China

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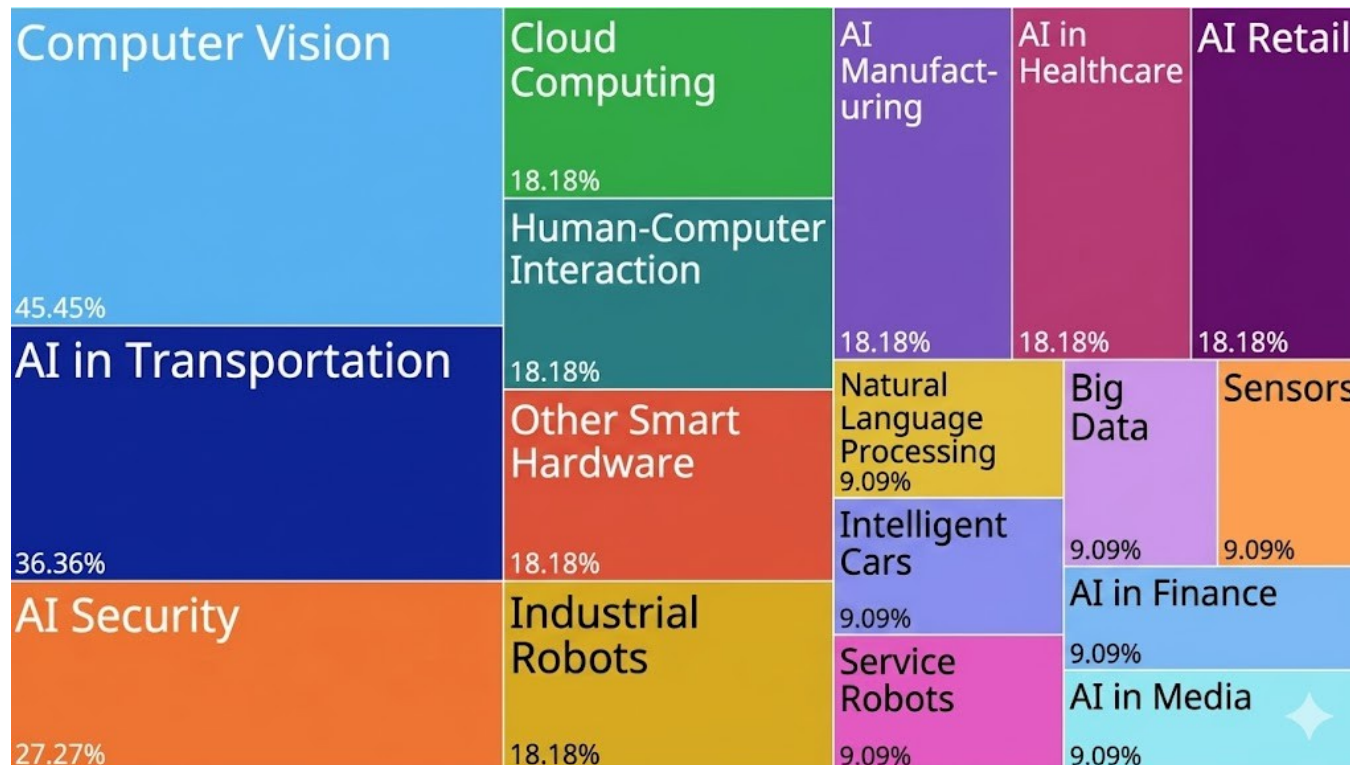
- Most Popular "Algorithm Engineer Position"



# CV in China

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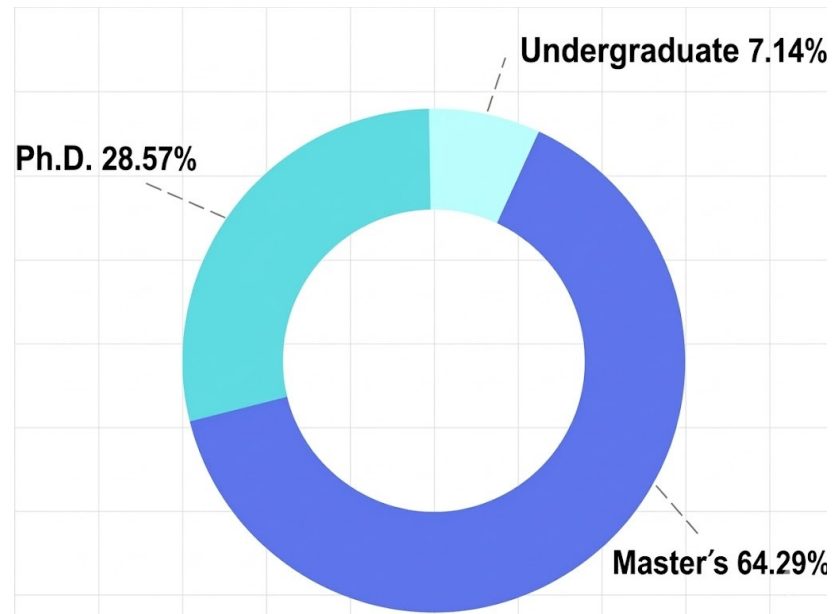
- Companies are involved in a diverse array of projects, placing extremely high demands on the versatility of talent.



# CV in China

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- Over 90% of enterprises require master's degree or above, preferring talent with diverse professional backgrounds

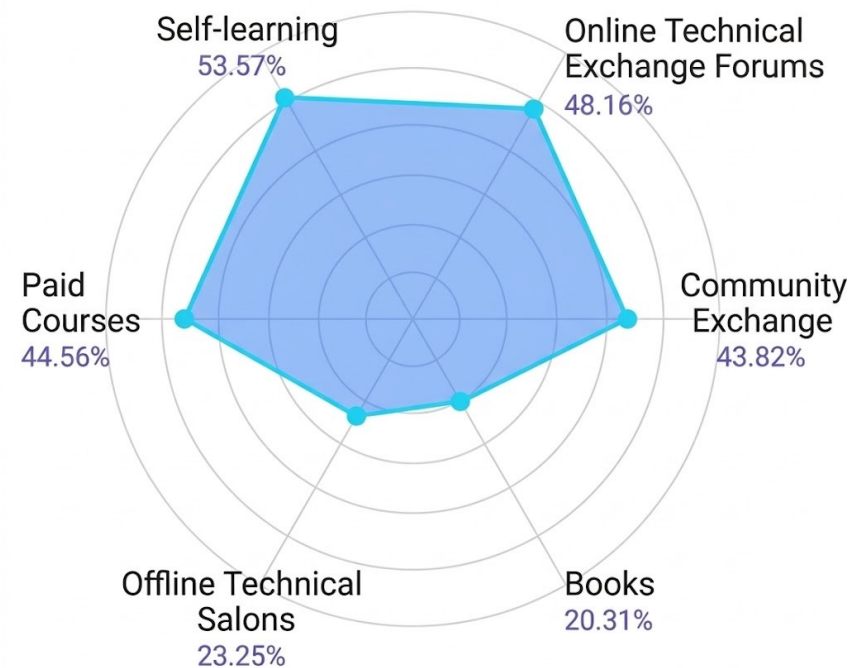




# CV in China

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- Computer vision talents mostly use their spare time to improve their technical skills



# Major Conferences and Publications

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- Three Major Conferences
  - International Conference on Computer Vision (ICCV) (biennial)
  - European Conference on Computer Vision (ECCV) (biennial)
  - Conference on Computer Vision and Pattern Recognition (CVPR) (annual)
- Two Major Journals
  - IEEE Trans PAMI
  - International Journal of Computer Vision
- One Important Award: Marr Prize

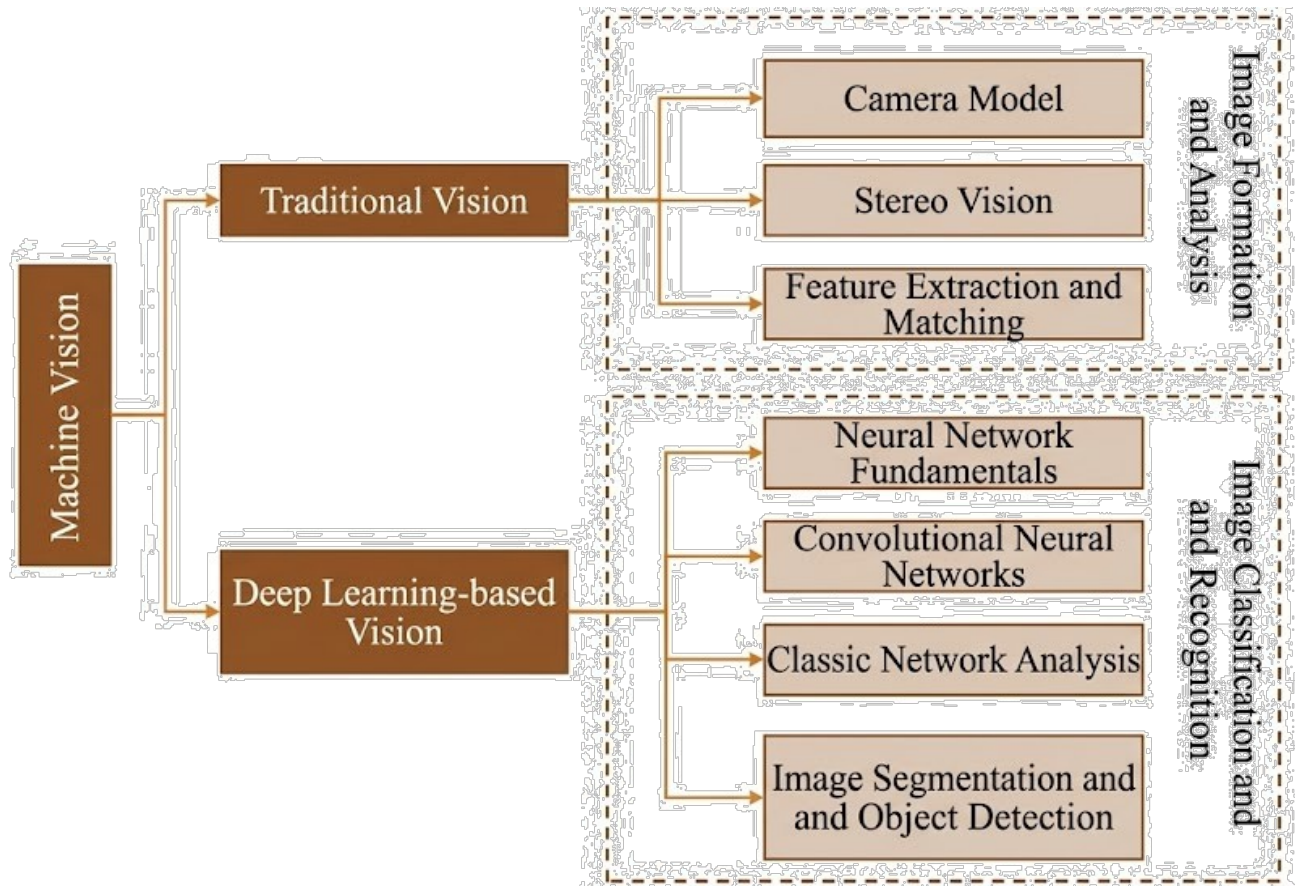
# Main Learning Resources

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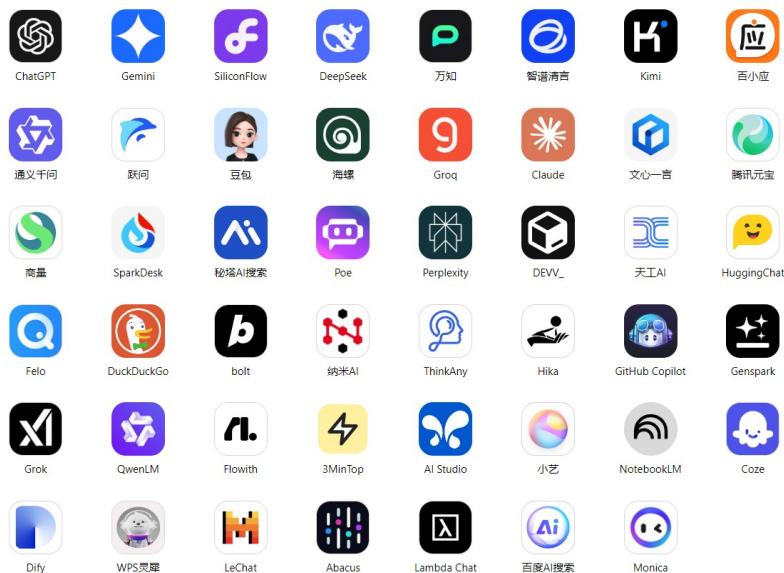
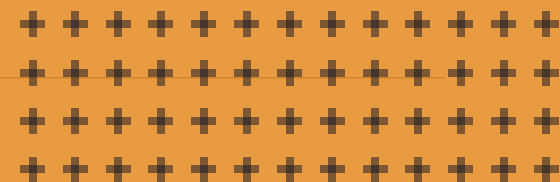
- CS131: Computer Vision: Foundations and Applications
  - Fall 2018, Juan Carlos Niebles and Ranjay Krishna
  - Undergraduate introductory class
- CS231a: Computer Vision, from 3D Reconstruction to Recognition
  - Professor Silvio Savarese
  - Image processing, cameras, 3D reconstruction, segmentation, object recognition, scene understanding; not just deep learning
- CS 230: Deep Learning
  - Spring 2019, Prof. Andrew Ng and Kian Katanforoosh
- CS231n: Convolutional Neural Networks for Visual Recognition
  - Justin Johnson & Serena Yeung & Fei-Fei Li
  - Focusing on applications of deep learning to computer vision

# Course Content

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# About the lecturer



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Contact Email: 2015019@hebut.edu.cn

Personal Homepage: <https://wot18.github.io>

Course Homepage: <https://github.com/wot18/Machine-Vision-for-Oversea-Students.git>

# Course Resources & Requirements

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- Textbooks:
  - 1. Szeliski, R. (2022). **Computer Vision: Algorithms and Applications** (2nd ed.). Springer.
  - The author of this book has long provided a free PDF draft version on their personal homepage.
  - Link: <https://szeliski.org/Book/>
  - 2. Zhang, A., Lipton, Z. C., Li, M., & Smola, A. J. (2023). **Dive into Deep Learning**. Cambridge University Press.
  - This is a continuously updated open-source interactive textbook that offers not only detailed theory but also executable code blocks.
  - Link: <https://d2l.ai/>
- Experimental Enviroments:
  - <https://aistudio.baidu.com/aistudio/education/group/invitation/37692?code=214Ac1>
- Exam format:
  - A report with codes and decriptions written by jupyternotebook.